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Toilet arrangement for swirl flushing

Abstract

A toilet bowl of a toilet, the toilet bowl comprising: a toilet base bowl comprising an upper portion and a toilet rim disposed in the upper portion of the toilet base bowl. The toilet rim comprises a tubular structure comprising a first end configured to receive water, a second end, and a middle section disposed between the first end and the second end. The water is configured to flow in a water flow direction towards the second end, and the second end is closed. The toilet rim also comprises a plurality of outlet conduits extending from the middle section directed towards an interior of the toilet base bowl. The plurality of outlet conduits is directed at an acute angle relative to the water flow direction, such that plurality of outlet conduits allow the water to enter the interior of the toilet base bowl in a swirl-flow manner.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit of priority to: Indian Patent Application No. 202211004621 filed in the Indian Intellectual Property Office on Jan. 27, 2022, which is hereby incorporated by reference in its entirety.

FIELD

(2) The present disclosure relates to a toilet bowl of a toilet arrangement. More particularly, the present disclosure relates to a toilet rim for the toilet bowl of the toilet arrangement.

BACKGROUND

(3) This section is intended to provide information relating to the field of the present disclosure and thus any approach or functionality described below should not be assumed to be qualified as prior art merely by its inclusion in this section.

(4) Toilet arrangements are known in sanitation industry, to enable persons to defecate in good sanitary conditions. A typical toilet arrangement includes a toilet bowl (also referred to as a ‘water closet’, interchangeably hereinafter) and a flush arrangement. The toilet bowl is provided to receive faecal (fecal) matter, when a user performs defecation thereon. The flush arrangement includes a flush tank and a flush valve unit. The flush tank stores water therein. The flush valve unit is fluidly connected between the flush tank and the toilet bowl, to allow/restrict a supply of water from the flush tank to the toilet bowl, upon actuation/release of at least one flush button. This results in flushing of faecal matter therefrom.

(5) The toilet bowl includes a toilet base bowl and a toilet rim. The toilet base bowl is formed of ceramic material, including a lower portion and an upper portion. The lower portion is shaped in form of a bowl, while the upper portion is shaped in form of an oval-shaped peripheral structure, which support the person while defaecating. Further, the toilet rim is arranged within the upper portion of the toilet base bowl, such that the toilet rim receives water from the flush tank and distributes the same in the toilet base bowl, for flushing.

(6) The toilet rim includes an oval-shaped closed loop tubular structure and a plurality of outlet conduits extending perpendicularly downwards from the oval-shaped closed loop tubular structure towards the toilet base bowl. Further, the oval-shaped closed loop tubular structure is fluidly connected to the flush arrangement, to receive water therefrom. In particular, the oval-shaped closed loop tubular structure includes a T-joint fluidly connected with a supply pipe extending from the flush arrangement. Accordingly, water, as received from the flush arrangement, is divided in two streams of water in the oval-shaped closed loop tubular structure, which is released through the plurality of outlet conduits to the toilet bowl. This reduces a water pressure released from the toilet rim to the toilet base bowl and thus requires more water to perform flushing. Further, in toilet rims, the plurality of outlet conduits extend perpendicularly downwards from the oval-shaped closed loop tubular structure. This results in release of water from the toilet rim to the toilet base bowl in a relatively turbulent manner, which may further cause requirement of relatively more water to perform flushing. Also, a flow velocity of water in the oval-shaped closed loop tubular structure reduces drastically between the T-joint and the point of meeting of the two streams of water, in particular, a flow velocity of water at a position in proximity of the T-joint is more than a flow velocity of water at a position of meeting of the two streams of water. This results in loss of potential energy of water and thus requires more water for flushing.

(7) Accordingly, in light of the aforementioned drawbacks and several other inherent in the arts, there is a well felt need to provide a toilet bowl of a toilet arrangement that uses relatively less water for flushing purposes.

SUMMARY

(8) This section is intended to introduce certain objects of the disclosed method and system in a simplified form and is not intended to identify the key advantages or features of the present disclosure.

(9) It is an object of the present disclosure to provide a toilet bowl of a toilet arrangement. The toilet bowl comprises a toilet base bowl and a toilet rim. The toilet rim is arranged with an upper portion of the toilet base bowl and comprises a tubular structure and a plurality of outlet conduits. The tubular structure defines a first end, a second end, and a middle section defined therebetween. The tubular structure is configured to receive water through the first end, to be flown in a water flow direction towards the second end, while the second end is closed. The plurality of outlet conduits extends from the middle section directed towards an interior of the toilet base bowl, the plurality of outlet conduits being angled at an acute angle relative to the water flow direction, such that a plurality of outlet conduits allow exit of water to an interior of the toilet base bowl, in a swirl-flow manner.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The present disclosure, both as to its organization and manner of operation, together with further objects and advantages, may be understood by reference to the following description, taken in connection with the accompanying drawings. These and other details of the present disclosure will be described in connection with the accompanying drawings, which are furnished only by way of illustration and not in limitation of the present disclosure, and in which drawings:

- (2) FIG. 1 is a perspective view of a toilet arrangement, in accordance with the concepts of the present disclosure.
- (3) FIG. 2 is a top view of a toilet bowl of the toilet arrangement of FIG. 1, in accordance with the concepts of the present disclosure.
- (4) FIG. 3 is a top view of a toilet rim of the toilet bowl of FIG. 2, in accordance with a first example of the present disclosure.
- (5) FIG. 4 is a side view of the toilet rim of the toilet bowl of FIG. 2, in accordance with the first example of the present disclosure.
- (6) FIG. 5 is a top view of the toilet rim of the toilet bowl, in accordance with the concepts of a second example of the toilet bowl disclosed in the present disclosure.
- (7) FIG. 6 is a side view of the toilet rim of the toilet bowl, in accordance with the concepts of the second example of the toilet bowl disclosed in the present disclosure.
- (8) FIG. 7 is a schematic of a section of a portion of the toilet bowl of FIG. 2, in accordance with the second example of the present disclosure.
- (9) FIG. 8 is a flow chart of a method for cleaning a toilet base bowl of a toilet by using a toilet rim according to an example of the present disclosure.

DETAILED DESCRIPTION

(10) In the following description, for the purposes of explanation, various specific details are set forth in order to provide a thorough understanding of embodiments of the present disclosure. It should be apparent, however, that embodiments of the present disclosure may be practiced without these specific details. Several features described hereafter can each be used independently of one another or with any combination of other features. An individual feature may not address any of the problems discussed above or might address only one of the problems discussed above. Some of the problems discussed above might not be fully addressed by any of the features described herein. Example embodiments of the present disclosure are described below, as illustrated in various drawings in which like reference numerals refer to the same parts throughout the different drawings.

(11) FIG. 1 shows a perspective view of a toilet arrangement **100**, in accordance with the concepts of the present disclosure. The toilet arrangement **100** facilitates a person to defecate in good sanitary conditions. The toilet arrangement **100** (e.g., a toilet) is generally known to include a toilet bowl **102** and a flush arrangement **104**.

(12) The toilet bowl **102** is a bowl structure, which is provided to receive faecal matter, when a user performs defecation thereon. Details of structure and arrangement of the toilet bowl **102** are explained below in details.

(13) The flush arrangement **104** is provided to supply water to the toilet bowl **102**, for flushing faecal matter therefrom. The flush arrangement **104** includes a flush tank **106** and a flush valve unit. The flush tank **106** is configured to store water therein. Further, the flush tank **106** of the flush arrangement **104** is fluidly connected to the toilet bowl **102** with use of a supply pipe **110**. The flush valve unit is positioned within the supply pipe **110** and is configured to allow/restrict the supply of water from the flush tank **106** of the flush arrangement **104** to the toilet bowl **102**. In particular, the flush valve unit allows/restrict the supply of water from the flush tank **106** of the flush arrangement **104** to the toilet bowl **102**, upon actuation/release of at least one flush button **112**. Flushing is enabled in the toilet bowl **102**, corresponding to the supply of water from the flush tank **106** of the flush arrangement **104** to the toilet bowl **102**. Although, the present disclosure describes the toilet arrangement **100** as a tank-based toilet arrangement, it may be apparent to a person having ordinary skill in the art that the toilet arrangement can also be an in-line toilet arrangement. A structure and an arrangement of the toilet bowl **102** are explained in detail hereinafter.

(14) FIG. 1 shows a perspective view of a toilet arrangement **100**, in accordance with the concept of the present disclosure. FIG. 2 shows a top view of the toilet bowl **102** of the toilet arrangement

100, in accordance with the concepts of the present disclosure. FIG. 1 and FIG. 2 should be referred to in conjunction with each other in order to clearly understand the concepts of the present disclosure related to the toilet bowl **102**. The toilet bowl **102** includes a toilet base bowl **114** and a toilet rim **116**. In a second example of the toilet bowl **102**, each of the toilet base bowl **114** and the toilet rim **116** is formed as a separate component and further connected together to form the toilet bowl **102**. In a first example of the toilet bowl **102**, each of the toilet base bowl **114** and the toilet rim **116** is integrally formed as a singular integrated component. Both the embodiments are within a scope of the present disclosure.

(15) Concept in respect of the first example of the toilet bowl **102** is explained hereinafter. In the first example of the toilet bowl **102**, the toilet base bowl **114** and the toilet rim **116** are integrally formed. In the first example, each of the toilet base bowl **114** and the toilet rim **116** are formed of ceramic material. In such embodiment, the toilet base bowl **114** and the toilet rim **116** are formed by casting in a single casting machine. The toilet base bowl **114** includes a lower portion **114b** and an upper portion **114a**. The toilet rim **116** is arranged with the upper portion **114a** of the toilet base bowl **114** and is configured to distribute water as received from the flushing arrangement **104**, within the toilet base bowl **114**, for enabling flushing of faecal matter therefrom.

(16) FIG. 3 is a top view of the toilet rim **116** of the toilet bowl **102**, in accordance with the concepts of the first example of the toilet bowl **102** disclosed in the present disclosure. FIG. 4 is a side view of the toilet rim **116** of the toilet bowl **102**, in accordance with the concepts of the first example of the toilet bowl **102** disclosed in the present disclosure. FIG. 3 and FIG. 4 should be referred to, in conjunction with each other, in order to clearly understand concepts of the present disclosure with respect to the first example of the toilet rim **116**. Although, FIG. 3 and FIG. 4, for illustrative purposes, show the toilet rim **116** as a separate component, it may be apparent to a person having ordinary skill in the art that the toilet rim **116** may be the integral component of the toilet bowl **102**, in accordance with the first example of the toilet bowl **102**. In particular, the toilet rim **116** includes a tubular structure **118** and a plurality of outlet conduits **120**.

(17) The tubular structure **118** is provided in form of a singular tubular structure, suitably bent to form an oval-structured unit, in particular an open-looped oval-structure unit, conforming to the upper portion of the toilet base bowl **114**. Although, the present disclosure is explained as the tubular structure **118** being deployed as the open-looped oval-structure unit, it may be apparent to a person having ordinary skill in the art that the tubular structure **118** can also be deployed as an open-looped square structured unit and/or an open-looped circular structured unit, depending on a shape of the upper portion **114a** of the toilet base bowl **114**. The tubular structure **118** defines or comprises a first end **118a**, a second end **118b**, and a middle section **118c** defined therebetween. The first end **118a** of the tubular structure **118** is fluidly connected to the supply pipe **110** of the flushing arrangement **104**, to receive water therefrom. Further, the second end **118b** of the tubular structure **118** is closed. With such an arrangement, the tubular structure **118** is configured to receive water through the first end **118a**, to be flown in a water flow direction towards the second end **118b** through the middle section **118c**. Furthermore, the tubular structure **118** is sequentially divided into four sections, namely a first section **118e**, a second section **118f**, a third section **118g**, and a fourth section **118h**, in the water flow direction. The first section **118e** includes the first end **118a**, and the fourth section **118h** includes the second end **118b**. An interior of the tubular structure **118** has a cross-section area that decreases continuously from the first end **118a** to the second end **118b** in the water flow direction. The interior of at least one portion of the tubular structure **118** has an elliptical cross-section area. This allows a smooth and laminar flow of water within the tubular structure **118**, while flowing from the first end **118a** to the second end **118b** in the water flow direction. In particular, the cross-sectional area of the at an end of the first section **118e** in the water flow direction is 50% of a cross-section area at the first end **118a** defined by the first section **118e** of the tubular structure **118**. Moreover, a cross-section area at an end of the second section **118f** in the water flow direction is 36% of the cross-section area at the first end **118a** defined by the first

section **118e** of the tubular structure **118**. Moreover, a cross-section area at an end of the third section **118g** in the water flow direction is 25% of the cross-section area of the first end **118a** defined by the first section **118e** of the tubular structure **118**. Furthermore, a cross-section area at the second end **118b** defined by the fourth section **118h** of the tubular structure in the water flow direction is 9.5% of the cross-section area of the first end **118a** defined by the first section **118e** of the tubular structure **118**. It may also be noted that the cross-section of the tubular structure **118** decreases continuously from the first end **118a** to the second end **118b**, in particular, the cross-section of the tubular structure **118** decreases continuously between the first section **118e**, the second section **118f**, the third section **118g**, and the fourth section **118h**. This results in positioning of the first end **118a** of the tubular structure **118** at a relatively higher height than the second end **118b** of the tubular structure **118**.

(18) The plurality of outlet conduits **120** are configured to release water received in the tubular structure **118** to the toilet base bowl **114**, for flushing. The plurality of outlet conduits **120** are relatively small conduit pipes extending from the middle section **118c** directed towards an interior of the toilet base bowl **114**. The plurality of outlet conduits **120** are angled or directed at an acute angle relative to the water flow direction. As the plurality of outlet conduits **120** are angled at an acute angle, the plurality of outlet conduits **120** allow exit of water to an interior of the toilet base bowl **114**, in a swirl-flow manner. Such exit of water to the interior of the toilet base bowl **114** in the swirl-flow manner requires relatively less water for flushing the faecal matter from the toilet base bowl **114**.

(19) Concepts in accordance with the second example of the toilet bowl **102**, will be explained in detail hereinafter. FIG. 5 is a top view of a schematic of the toilet bowl **102**, in accordance with the concepts of the second example of the toilet bowl **102** disclosed in the present disclosure. FIG. 6 is a side view of a schematic of the toilet bowl **102**, in accordance with the concepts of the second example of the toilet bowl **102** disclosed in the present disclosure. FIG. 7 shows an isometric view of a portion of a schematic of the toilet bowl **102**, in accordance with the second example of the toilet bowl **102**. FIG. 5, FIG. 6, and FIG. 7, should be referred to in conjunction with each other, in order to clearly understand the concepts of the second example of the toilet bowl **102** disclosed in the present disclosure. In the second example of the toilet bowl **102**, the toilet base bowl **114'** and the toilet rim **116'** are separately formed. In the second example of the toilet bowl **102**, the toilet base bowl **114'** is formed of a ceramic material, while the toilet rim **116'** is at least partially formed of an elastomer material.

(20) A structure and an arrangement of the toilet base bowl **114'** in accordance with the second example of the toilet bowl **102** are similar to a structure and arrangement of the toilet base bowl **114'** in accordance with the first example of the toilet bowl **102**. Particularly, the toilet base bowl **114'** includes a lower portion **114b'** and an upper portion **114a'**. In one embodiment, the upper portion **114a'** forms a step relative to the lower portion **114b'** of the toilet base bowl **114'**. In another embodiment, the upper portion **114a'** forms a U-shaped cavity **114d'** therein. Concepts of the present disclosure hereinafter are explained in relation to mounting of the toilet rim **116'**, in the U-shaped cavity **114d'** in the upper portion **114a'** of the toilet base bowl **114'**. Similar concepts of the present disclosure may be envisioned to be applied in relation to mounting of the toilet rim **116'**, on the step of the upper portion **114a'** of the toilet base bowl **114'**.

(21) The toilet rim **116'** is at least partially formed of an elastomer material and is positioned within the U-shaped cavity **114d'** formed in the upper portion **114a'** of the toilet base bowl **114'**. The toilet rim **116'** includes a tubular structure **118'**, a connection adapter **119'**, and a plurality of outlet conduits **120'**.

(22) The tubular structure **118'** is formed of a soft elastomer material, such as but not limited to, thermoplastic elastomer (TPE), rubber, EVA (Elastic Plastic), foam, and the like. The tubular structure **118'** is provided in form of a singular tubular structure, suitably bent to form an oval-structured unit, in particular an open-looped oval-structure unit, conforming to or corresponding to

the upper portion of the toilet base bowl **114'**. Although, the present disclosure is explained as the tubular structure **118'** being deployed as the open-looped oval-structure unit, it may be apparent to a person having ordinary skill in the art that the tubular structure **118'** can also be deployed as an open-looped square structured unit, and/or an open-looped circular structured unit, depending on a shape of the upper portion **114a'** of the toilet base bowl **114'**. The tubular structure **118'** defines a first end **118a'**, a second end **118b'**, and a middle section **118c'** defined therebetween. Further, the second end **118b'** of the tubular structure **118'** is closed. Furthermore, the tubular structure **118'** is sequentially divided into four sections, namely a first section **118e'**, a second section **118f'**, a third section **118g'**, and a fourth section **118h'**, in the water flow direction. The first section **118e'** includes the first end **118a'**, and the fourth section **118h'** includes the second end **118b'**. An interior of the tubular structure **118'** has a cross-section area that decreases continuously from the first end **118a'** to the second end **118b'** in the water flow direction. The interior of at least one portion of the tubular structure **118'** has an elliptical cross-section area. This allows a smooth and laminar flow of water within the tubular structure **118'**, while flowing from the first end **118a'** to the second end **118b'** in the water flow direction. In particular, the cross-sectional area of the at an end of the first section **118e'** in the water flow direction is 50% of a cross-section area at the first end **118a'** defined by the first section **118e'** of the tubular structure **118'**. Moreover, a cross-section area at an end of the second section **118f'** in the water flow direction is 36% of the cross-section area at the first end **118a'** defined by the first section **118e'** of the tubular structure **118'**. Moreover, a cross-section area at an end of the third section **118g'** in the water flow direction is 25% of the cross-section area of the first end **118a'** defined by the first section **118e'** of the tubular structure **118'**. Furthermore, a cross-section area at the second end **118b'** defined by the fourth section **118h'** of the tubular structure in the water flow direction is 9.5% of the cross-section area of the first end **118a'** defined by the first section **118e'** of the tubular structure **118'**. It may also be noted that the cross-section of the tubular structure **118'** decreases continuously from the first end **118a'** to the second end **118b'**, in particular, the cross-section of the tubular structure **118'** decreases continuously between the first section **118e'**, the second section **118f'**, the third section **118g'**, and the fourth section **118h'**. This results in positioning of the first end **118a'** of the tubular structure **118'** at a relatively higher height than the second end **118b'** of the tubular structure **118'**.

(23) The connection adapter **119'** is provided to fluidly couple the tubular structure **118'** to the supply pipe **110** and is made up of a relatively hard elastomer material, such as but not limited to, thermoplastic elastomer (TPE), rubber, EVA (Elastic Plastic), foam, and the like. For such purposes, the connection adapter **119'** is fluidly coupled to the first end **118a'** of the tubular structure **118'**. Further, the connection adapter **119'** is also fluidly coupled to the supply pipe **110**. With such arrangement, the tubular structure **118'** may receive water through the connection adapter **119'** at the first end **118a'**, to be flown in a water flow direction towards the second end **118b'** through the middle section **118c'**.

(24) The plurality of outlet conduits **120'** are configured to release water received in the tubular structure **118'** to the toilet base bowl **114'**, for flushing. The plurality of outlet conduits **120'** are relatively small conduit pipes extending from the middle section **118c'** directed towards an interior of the toilet base bowl **114'**. The plurality of outlet conduits **120'** are angled at an acute angle relative to the water flow direction. As the plurality of outlet conduits **120'** are angled at an acute angle, the plurality of outlet conduits **120'** allow exit of water to an interior of the toilet base bowl **114'**, in a swirl-flow manner. Such exit of water to the interior of the toilet base bowl **114'** in the swirl-flow manner, requires relatively less water for flushing the faecal matter from the toilet base bowl **114'**. For attachment of the toilet rim **116'** to the toilet base bowl **114'**, the tubular structure **118'** of the toilet rim **116'** is attached within the U-shaped cavity **114d'** formed within the upper portion **114a'** of the toilet base bowl **114'**. The tubular structure **118'** of the toilet rim **116'** is attached within the U-shaped cavity **114d'** formed within the upper portion **114a'** of the toilet base bowl **114'**, via an attachment means. The attachment means includes, but is not limited to, an

adhesive attachment, a C-clip attachment, a plastic attachment, a bolt attachment, and/or any other similar attachment means. With such arrangement, the tubular structure **118'** may receive water through the connection adapter **119'** at the first end **118a'** of the tubular structure **118'**, to be flown in a water flow direction towards the second end **118b'**, and further exit the water to the interior of the toilet base bowl **114'** through the plurality of outlet conduits **120'**.

(25) Various advantages of the toilet bowl **102**, as disclosed in the present disclosure, exists. For example, as the plurality of outlet conduits **120, 120'** of the toilet rim **116, 116'** of the toilet bowl **102** are angled at an acute angle, the plurality of outlet conduits **120, 120'** allow exit of water to an interior of the toilet base bowl **114, 114'**, in a swirl-flow manner. Such exit of water to the interior of the toilet base bowl **114, 114'** in the swirl-flow manner, requires relatively less water for flushing the faecal matter from the toilet base bowl **114, 114'**. Additionally, such exit of water to the interior of the toilet base bowl **114, 114'** in the swirl-flow manner causes a relatively more coverage of flush area within the interior of the toilet base bowl **114, 114'** and thus causes better cleaning. Another advantage of the present disclosure relates to water attachment of the tubular structure **118, 118'** at the first end **118a, 118a'**, and the second end **118b, 118b'** being closed. This enables flow of water only in one direction forming only one stream of water, instead of the T-joint. This avoids any backflow or cancellation of flow of water, and thus saves a kinetic energy of water, and consequently requires less water for flushing operations. Yet another advantage of the toilet bowl **102**, as disclosed in the present disclosure, relates to the first end **118a, 118a'** of the tubular structure **118, 118'** been placed at a relatively higher height than the second end **118b, 118b'** of the tubular structure **118, 118'**. This results in a relatively increased average velocity of water exiting from the tubular structure **118, 118'** of the toilet rim **116, 116'** to the toilet base bowl **114, 114'** and thus requires relatively less water, for flushing operations. Yet another advantage of the present disclosure relates to the continuously decreasing cross-section of the tubular structure **118, 118'** between the first section **118e, 118e'**, the second section **118f, 118f'**, the third section **118g, 118g'**, and the fourth section **118h, 118h'**. This facilitates in gradual decrease in average velocity of water flowing in the tubular structure **118, 118'**. Thus, the kinetic energy of the water flowing in the tubular structure **118, 118'** is conserved and adds to low requirements of water for flushing purposes. As the interior of at least one portion of the tubular structure **118, 118'** has an elliptical cross-section area, it allows a smooth and laminar flow of water within the tubular structure **118, 118'**, while flowing from the first end **118a, 118a'** to the second end **118b, 118b'** in the water flow direction. Overall, the specific design, structure, and arrangement of the toilet rim **116, 116'** of the toilet bowl **102**, as disclosed in the present disclosure, requires relatively less water, for flushing purposes.

(26) The present disclosure also provides a toilet arrangement **100** (e.g., a toilet). The toilet arrangement **100** comprises the toilet base bowl **114** and the toilet rim **116** according to any examples as described above.

(27) In an embodiment, the toilet arrangement **100** includes a base (e.g., a pedestal, bowl, etc.) and the flush tank **106**. The base is configured to be attached to another object such as a drainpipe, floor, or another suitable object. The base includes a bowl, a sump (e.g., a receptacle) disposed below the bowl, and a trapway fluidly connecting the bowl to a drainpipe or sewage line. The flush tank **106** may be supported by the base, such as an upper surface of a rim. The flush tank **106** may be integrally formed with the base as a single unitary body. In other embodiments, the flush tank **106** may be formed separately from the base and coupled (e.g., attached, secured, fastened, connected, etc.) to the base. The toilet arrangement **100** may further include a tank lid covering an opening and inner cavity in the tank. The toilet arrangement **100** may include a seat assembly including a seat and a seat cover rotatably coupled to the base. The toilet arrangement **100** may further include a hinge assembly.

(28) In another embodiment, the toilet arrangement **100** may be a tankless toilet. The toilet arrangement **100** includes a base and a seat assembly coupled to the base. The base includes a

bowl, a sump disposed below the bowl, and a trapway fluidly connecting the bowl to a drainpipe or sewage line. The toilet arrangement **100** includes a waterline that supplies the toilet arrangement **100** with water. The toilet arrangement **100** may further include a seat assembly including a seat and a seat cover rotatably coupled to the base. The toilets described above are provided herein as non-limiting examples of toilets that may be configured to utilize aspects of the present disclosure. (29) In some examples, a bidet may be included in a seat or pedestal of a toilet. In other examples, the bidet may be manufactured separately from and attached or coupled to a seat or pedestal of a toilet. The bidet includes a housing. The housing is configured to receive a flow of water through a housing inlet and dispense the flow of water from a housing outlet. The housing inlet and housing outlet may be located on opposite ends of the housing from one another, such that water may flow through the housing from the housing inlet to the housing outlet. In some examples, the housing further includes a chamber. As the housing receives the flow of water, the chamber may fill with water and provide a flow of water between the housing inlet and the housing outlet. The chamber may be configured to contain the flow of water and direct the flow of water from the housing inlet to the housing outlet. After the chamber has filled with water, the flow of water may travel along a substantially linear path between the housing inlet and the housing outlet. In some examples, one or more walls within the housing may be included to help direct a flow of water between the housing inlet and the housing outlet. The bidet may further include a housing inlet conduit configured to direct a flow of water to the housing inlet. The housing inlet conduit may be coupled to a water supply such as tank or waterline. The housing may further include a gear assembly or a portion of the gear assembly.

(30) FIG. **8** is a flow chart of a method for cleaning a toilet base bowl of a toilet by using a toilet rim according to an example of the present disclosure. The toilet rim used in the method may be toilet rim **116** according to any of the foregoing examples and may be configured to perform an operation, function, or the like as described in the present disclosure.

(31) At act **S101**, an actuation of the flush button **112** may open the flush valve unit to discharge water from the flush tank **106**. As noted above, the flush valve unit is positioned within the supply pipe **110** and is configured to allow/restrict the supply of water from the flush tank **106** of the flush arrangement **104** to the toilet bowl **102**. In particular, the flush valve unit allows/restrict the supply of water from the flush tank **106** of the flush arrangement **104** to the toilet bowl **102**, upon actuation/release of at least one flush button **112**.

(32) At act **S102**, the supply pipe **110** may deliver the water from the tank **106** to the first end **118a** of the toilet rim **116** disposed in the upper portion **114a** of the toilet base bowl **114**. As noted above, the first end **118a** is fluidly connected to the supply pipe **110** of the flushing arrangement **104** to receive the water from the flushing arrangement **104**.

(33) At act **S103**, the toilet rim **116** may deliver the water from the first end **118a** of the toilet rim **116** in the water flow direction towards the second end **118b** of the toilet rim **116**. As noted above, the second end **118b** is closed. With such an arrangement, the tubular structure **118** is configured to receive water through the first end **118a**, to be flown in the water flow direction towards the second end **118b** through the middle section **118c**.

(34) At act **S104**, the plurality of outlet conduits directed towards an interior of the toilet base bowl **114** may deliver the water in the toilet rim **116** to the interior of the toilet base bowl **114** in a swirl-flow manner. As noted above, the plurality of outlet conduits **120** are relatively small conduit pipes extending from the middle section **118c** directed towards the interior of the toilet base bowl **114**. The plurality of outlet conduits **120** are angled or directed at the acute angle relative to the water flow direction. As the plurality of outlet conduits **120** are angled at the acute angle, the plurality of outlet conduits **120** allow exit of water to the interior of the toilet base bowl **114** in the swirl-flow manner.

(35) Although particular embodiments have been disclosed herein in detail, this is for illustrative purposes only and is not intended in any way to limit the intended scope of the present disclosure.

Variations and adaptations of the system and the method as described herein do not depart from the spirit and scope of the present disclosure and should be within the expertise of a person having ordinary skill in the art.

LIST OF COMPONENTS

(36) **100**—Toilet Arrangement **102**—Toilet Bowl **104**—Flushing Arrangement **106**—Flush Tank **110**—Supply Pipe **112**—Flush Button **114, 114'**—Toilet Base Bowl **116, 116'**—Toilet Rim **114a, 114a'**—Upper Portion of **114 114b, 114b'**—Lower Portion of **114 116**—Toilet Rim **118, 118'**—Tubular Structure **118a, 118b**—First End, Second End of **118 118a', 118b'**—First End, Second End of **118' 118c**—Middle Section of **118 118c'**—Middle Section of **118' 119**—Connection Adapter **120**—Outlet Conduits

Claims

1. A toilet bowl of a toilet, the toilet bowl comprising: a toilet base bowl comprising an upper portion; and a toilet rim disposed in the upper portion of the toilet base bowl, the toilet rim comprising: a tubular structure comprising a first end configured to receive water, a second end, and a middle section disposed between the first end and the second end, wherein the water is configured to flow in a water flow direction towards the second end, and the second end is closed, wherein an interior of the tubular structure has a cross-section area that decreases continuously from the first end to the second end in the water flow direction; and a plurality of outlet conduits extending from the middle section directed towards an interior of the toilet base bowl, the plurality of outlet conduits being directed at an acute angle relative to the water flow direction, such that plurality of outlet conduits allow the water to enter the interior of the toilet base bowl in a swirl-flow manner.
2. The toilet bowl according to claim 1, wherein the tubular structure comprises a first section including the first end, a second section, a third section, and a fourth section including the second end.
3. The toilet bowl according to claim 2, wherein a cross-section area at an end of the first section in the water flow direction is 50% of a cross-section area at the first end included in the first section of the tubular structure.
4. The toilet bowl according to claim 2, wherein a cross-section area at an end of the second section in the water flow direction is 36% of the cross-section area at the first end included in the first section of the tubular structure.
5. The toilet bowl according to claim 2, wherein a cross-section area at an end of the third section in the water flow direction is 25% of the cross-section area at the first end included in the first section of the tubular structure.
6. The toilet bowl according to claim 2, wherein a cross-section area at the second end included in the fourth section of the tubular structure in the water flow direction is 9.5% of the cross-section area at the first end included in the first section of the tubular structure.
7. The toilet bowl according to claim 1, wherein the first end of the tubular structure of the toilet rim is configured to be fluidly connected to a flushing arrangement to receive the water from the flushing arrangement.
8. The toilet bowl according to claim 1, wherein the tubular structure of the toilet rim forms an oval-structured unit configured to be disposed in an inner periphery of the upper portion of the toilet base bowl.
9. The toilet bowl according to claim 1, wherein the first end of the tubular structure is disposed at a relatively higher level than the second end of the tubular structure.
10. The toilet bowl according to claim 1, wherein an interior of at least one portion of the tubular structure has an elliptical cross-section area.
11. A toilet rim for a toilet bowl of a toilet, the toilet rim comprising: a tubular structure comprising

a first end, a second end, and a middle section disposed between the first end and the second end; a connection adapter fluidly coupled to the first end of the tubular structure, wherein the tubular structure is configured to receive water through the connection adapter, the water is configured to flow in a water flow direction towards the second end, and the second end is closed, wherein an interior of the tubular structure has a cross-section area that decreases continuously from the first end to the second end in the water flow direction; and a plurality of outlet conduits extending from the middle section directed towards an interior of a toilet base bowl, the plurality of outlet conduits being directed at an acute angle relative to the water flow direction, such that plurality of outlet conduits allow the water to enter the interior of the toilet base bowl in a swirl-flow manner.

12. The toilet rim according to claim 11, wherein the tubular structure of the toilet rim is configured to be attached to a step formed on an upper portion of the toilet base bowl via an attachment means.

13. The toilet rim according to claim 11, wherein the tubular structure comprises a first section including the first end, a second section, a third section, and a fourth section including the second end.

14. The toilet rim according to claim 11, wherein the cross-section area at an end of the first section in the water flow direction is 50% of the cross-section area at the first end included in the first section of the tubular structure.

15. The toilet rim according to claim 11, wherein the cross-section area at an end of the second section in the water flow direction is 36% of the cross-section area at the first end included in the first section of the tubular structure.

16. The toilet rim according to claim 11, wherein the cross-section area at an end of the third section in the water flow direction is 25% of the cross-section area at the first end included in the first section of the tubular structure.

17. The toilet rim according to claim 11, wherein the cross-section area at the second end included in the fourth section of the tubular structure in the water flow direction is 9.5% of the cross-section area at the first end included in the first section of the tubular structure.

18. A method for cleaning a toilet base bowl of a toilet by using a toilet rim, the method comprising: opening a flush valve to discharge water from a tank in response to an actuation of a flush button; delivering, by a supply pipe, the water from the tank to a first end of a toilet rim disposed in an upper portion of the toilet base bowl; delivering, by the toilet rim, the water from the first end of the toilet rim in a water flow direction towards a second end of the toilet rim, wherein an interior of the tubular structure has a cross-section area that decreases continuously from the first end to the second end in the water flow direction; and delivering, by a plurality of outlet conduits directed towards an interior of the toilet base bowl, the water in the toilet rim to the interior of the toilet base bowl in a swirl-flow manner, wherein the plurality of outlet conduits extend from a middle section of the toilet rim and directed at an acute angle relative to the water flow direction, wherein the middle section of the toilet rim is disposed between the first end of the toilet rim and the second end of the toilet rim, and wherein the second end of the toilet rim is closed.
